

Structural Change in Malawi since Independence: Consequences of a Development Strategy Based on Large-scale Agriculture

JONATHAN KYDD

Chancellor College, University of Malawi, Limbe, Malawi

and

ROBERT CHRISTIANSEN

Colby College, Waterville, Maine

Summary. – This paper investigates the nature of the structural change which has occurred in the Malawian economy since independence and identifies key policies pursued by the government which it is argued have been decisive in shaping this development. **The salient feature of structural change is shown to be the rapid growth of large-scale agriculture, associated with an equally rapid transfer of labour into wage employment and a decline in the importance of peasant production. The implications of these changes for living standards and future development and policy are considered.**

1. INTRODUCTION

The purpose of this paper is to investigate structural change in the Malawian economy since political independence in 1964. We focus on three aspects of change. Firstly, we comment on some aspects of colonial economic development which are particularly relevant for an appreciation of what has happened since. Next, we document the structural changes of the last two decades with an emphasis on changes within the agricultural sector. Finally, we consider the consequences of the government's development strategy for current living standards and its implications for future economic development and policy.

2. RECENT LITERATURE

The literature on the development of the Malawian economy after Independence is remarkably small. For our purposes the important works are Humphrey (1975), Thomas (1975) and Acharya (1978, 1981a, b).¹ Both Humphrey and Thomas were writing at a time when the direction of trends in the Malawian economy was less clear than is the case today.

Humphrey was concerned with identifying 'economic progress and prospects'. While noting the strong growth of GDP since Independence he expresses concern over trends which he perceives in, firstly, the distribution of the gains of this growth and, secondly, a lack of government attention to distributional objectives. Another area of concern for Humphrey, rightly as it turns out, was the growing level of external debt contracted by Malawi. Thomas also mentions a number of areas of concern. Firstly, he notes the relatively poor performance of peasant agriculture in the context of the overall growth of the economy. Secondly, he notes the emerging policy bias in favour of large-scale agriculture, in particular the use of marketing board surpluses for financing the development of commercial agriculture. Thirdly, he notes the massive emphasis in government development expenditure on infrastructure, at the expense of health and education. Finally, Thomas shared Humphrey's concern over increasing inequity in distributional trends.

While both Humphrey and Thomas recognize the successful aspects of Malawi's development, essentially strong GDP growth without the benefit of mineral resources or a significant manufacturing base, this is balanced by an ack-

nowledgement of serious problems. In contrast, three recent papers by Acharya have tended to present economic development in Malawi in a singularly favourable light. Acharya's primary objective is to identify alternative development strategies and their consequences in post-colonial sub-Saharan Africa. Malawi enters the discussion, along with Kenya and the Ivory Coast, as an example of a successful 'market-oriented' development strategy.² Acharya compares 'market-oriented' strategies with the 'étatist' strategies which have been more common in post-colonial Africa. For Acharya, key facets of a market-oriented strategy are a lesser degree of government intervention in the economy allied to a willingness to let world prices determine domestic resource allocation and, consequently, a less pronounced bias against agriculture. Acharya sees price incentives for peasant production as the crucial component of a market-oriented strategy, which leads not only to strong growth in agricultural exports, but also to satisfactory development of the manufacturing sector and, he suggests, relatively equitable distribution of income.³ It is in this context that Acharya portrays Malawi as a market-oriented state which 'acted on the need to sustain remunerative price incentives for peasant agriculture'.⁴ Further, with respect to agricultural export performance, Acharya states that:

Ivory Coast and Malawi and, to a lesser extent, Kenya, chose to increase their specialization in agricultural exports together with some diversification into new agricultural commodities. It was no coincidence that these were also the countries which sustained the most favourable environment for private peasant agriculture.⁵

We do not take issue with Acharya's emphasis on the benefits of price incentives. However, one consequence of this paper is to demonstrate that Malawi's development has been far from the considerable success story which Acharya depicts. Indeed, any 'success' that Malawi has enjoyed has been seriously marred by declining relative and real living standards for the majority of the population and a sharp decline in the productive potential of peasant agriculture. As our analysis makes clear, these unfavourable developments are in part attributable to the fact that economic policy in Malawi has been much less market-oriented than Acharya would have us believe. While Malawi's development strategy is an example of a relatively low degree of policy bias against the agricultural sector as a whole, we argue later that government intervention to promote the growth of large-scale (estate) agriculture has

seriously damaged the development of peasant agriculture.

3. THE ECONOMY UP TO INDEPENDENCE IN 1964

The pre-colonial economy of Malawi was characterized by societies engaged largely in subsistence agriculture. These societies varied considerably with respect to their external contacts, from those engaging in limited local exchange of commodities such as iron, salt and food to societies which were substantially engaged in long-distance trade. The establishment of the British Protectorate in 1891 marked a new stage in the participation of societies within Malawi in international markets. Economic development in the colonial period was characterized by conflict between estate (plantation) agriculture and peasant agriculture. In addition, there were forces which led to the involvement of Malawians in international labour migration. For this reason, it is convenient to consider the changes wrought by colonial rule in terms of the evolution of three stylized subeconomies. These are the plantation or estate economy, the peasant cash cropping economy and the labour reserve economy.

The plantation subeconomy was the first to develop and the colonial administration encouraged it by facilitating the transfer of land to Europeans and by forcing Africans to seek wage employment in the estates through the imposition of the hut tax and other devices. By the turn of the century, the plantations, which were primarily engaged in coffee growing, had a value of output greater than the settler farms of Kenya or Southern Rhodesia.⁶ The strong growth of the plantation sector came to an end in 1902 as a result of a collapse in coffee prices caused by expanded Brazilian production, as well as increasing coffee disease problems in Nyasaland. With the failure of coffee, the estate sector turned to cotton and tobacco and, in the limited areas where the climate was suitable, tea. The sector grew slowly through to the end of the 1920s but never really prospered due to high transport costs, undercapitalization and the lack of technical skills and the short-term perspective of most European planters.⁷ The estate sector was dealt a severe blow by the collapse of cotton and tobacco prices in the early 1930s. Many of the tobacco estates were abandoned and were not to be redeveloped until the tobacco boom of the 1970s. From the early 1930s to Independence the impression is of a plantation sector which was

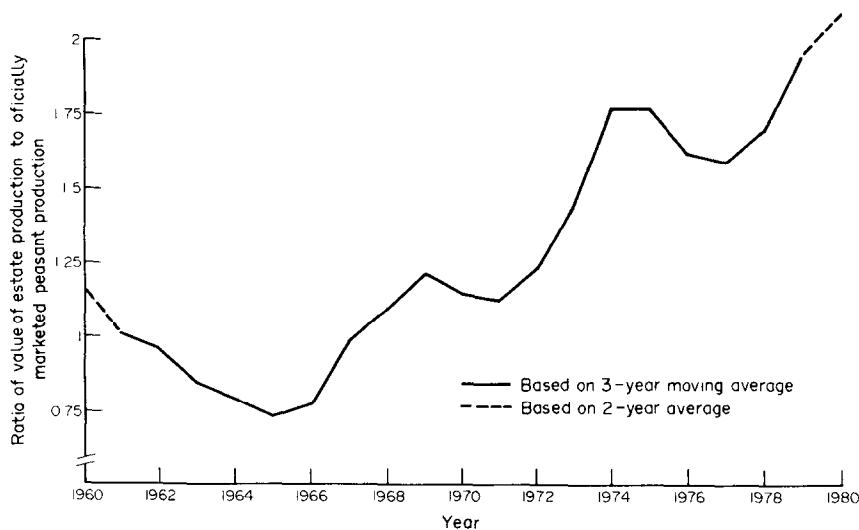


Figure 1. Ratio of value of estate production to value of officially marketed peasant production.

Statistics required for the preparation of Figure 1 (the ratio of the value of estate production to the value of officially marketed peasant production)*

Year	Ratio	Year	Ratio
1960	(1.15)	1971	1.12
1961	1.03	1972	1.23
1962	0.96	1973	1.44
1963	0.84	1974	1.76
1964	0.79	1975	1.76
1965	0.73	1976	1.67
1966	0.77	1977	1.58
1967	0.98	1978	1.68
1968	1.09	1979	1.93
1969	1.21	1980	(2.07)
1970	1.14		

Source: Calculated from estimates in Kydd and Christiansen (1981b), Appendix 9.

* Figures are calculated from 3-year moving averages with the exception of bracketed figures which are 2-year averages.

broadly static in terms of output and technology. The tea industry was a partial exception to this picture in that production grew significantly, by about 4% per annum in the 1940s and 1950s.

In the unpromising commercial environment in which the estates found themselves their survival required a search for the cheapest possible labour. The first instrument for obtaining labour, the hut tax, did not entirely meet the needs of the estates since adult men had the option of meeting tax obligations by growing cash crops or migration. Estate interests had only limited success in persuading the administration to close the cash cropping and migration

options. A solution to the 'labour problem' was finally achieved through settler control of land. Typical estates consisted of large tracts of land of which only a small proportion were used for growing cash crops. There was thus scope for offering land to peasant tenants who were mainly Lomwe people who had migrated into Nyasaland from Mozambique to avoid an even more oppressive Portuguese colonialism. The predominant form of tenancy, *thangata*, was a quasi-feudal arrangement under which peasants had to supply labour to the plantation in exchange for the use of a plot of land. Another form of tenancy which developed at about the same time as *thangata* is what is presently called

the 'visiting-tenant' system. This essentially was a share-cropping arrangement under which peasant families obtained land on estates on the condition that they grew a cash crop which was sold to the estate at a price determined by the estate owner. Discontent with *thangata* was an important factor in rural unrest in 1915 and in the early 1950s and *thangata* was abolished in the final years of the colonial period. The 'visiting-tenant' system, which had been less common than *thangata*, remained in operation in some areas and has grown rapidly in the 1970s in line with the development of burley tobacco production. As this suggests, the estate boom of the 1970s was characterized by the same need for low-cost labour as was earlier estate production.

The coming of the colonial administration and the development of the estates served as a stimulus for peasant production and sale of food surpluses. The dependence of the estate sector on food supplies from the peasant sector represents a crucial linkage which has endured to the present day.⁸ However, the development of peasant production of export crops resulted in conflict with estate interests not only for land and labour, but also for the support of a colonial administration anxious for some form of economic development.⁹ In the first decade of colonial rule the administration was unambiguous in its support of estate interests. However the setbacks experienced by the estate sector caused administration policy to move from the cautious sanctioning of peasant production of export crops in areas where there would be no sharp competition for resources with the estates to one of intervention to promote widespread peasant production. This emphasis resulted in the dominance of peasant agriculture from the late 1920s through to the mid-1960s. As Figure 1 shows, the dominance of the peasant sector began to recede in the late 1960s.

Given the demand for labour created by the development of mining and European commercial agriculture in South Africa and Rhodesia significant labour migration from Malawi was likely to have occurred. However, migration was encouraged by the fact that both estate and peasant cash cropping agriculture were confined to limited areas of the country, mainly the Southern Region and parts of the Central Region. As early as the late 1890s significant numbers of men responded to this external demand for labour and in 1900 estate interests petitioned for control of migration as a result of a labour shortage.¹⁰ Over the next four

decades, government policy on migration vacillated under the contrary pressures of, on the one hand, estate and missionary influences and, on the other, the benefits to government revenue from migrants' remittances. However, government policy was of limited relevance as independent migration was difficult to control and the authorities in the recipient territories were rarely co-operative. The growing importance of labour migration can be seen from the figures in Table 1, which are calculated from Boeder's (1974) estimates.¹¹ Labour migration peaked in the early 1970s and the 1977 Census Report shows that large numbers of migrants returned to Malawi during a period when opportunities for new migration were severely constrained. This reduction in labour migration was due, firstly, to a decrease in demand in neighbouring countries due to such factors as technical changes allied to changes in employment policy in South Africa, civil war in Zimbabwe and poor economic performance in Zambia and, secondly, to supply constraints imposed by the Malawian government. **An approximate estimate based on census data suggests that in 1977 the proportion of the population abroad as a percentage of the *de facto* population had fallen to 5%.¹² The curtailment of labour migration coincided with the increased demand for wage labour which accompanied the estate boom of the 1970s.¹³**

The impact of the changes in international migration patterns on population sex ratios and labour force composition is shown in Figure 2. Figure 2 shows clearly that there have been changes not only in the relative numbers of migrants, but also in the pattern of migration. There has been a great reduction in the frequency of longer-term migration, but, at the same time, significant numbers of young men (aged 20–24) are still engaging for short-term mine labour contracts.

Table 1. *Malawians abroad as a percentage of estimated de facto population of Malawi*

Year	Percentage abroad
1903	1.1
1911	2.5
1924	2.5
1938	6.8
1948	5.6
1958	6.2
1972	10.3

Source: Calculated from Boeder (1974), Appendices B and C.

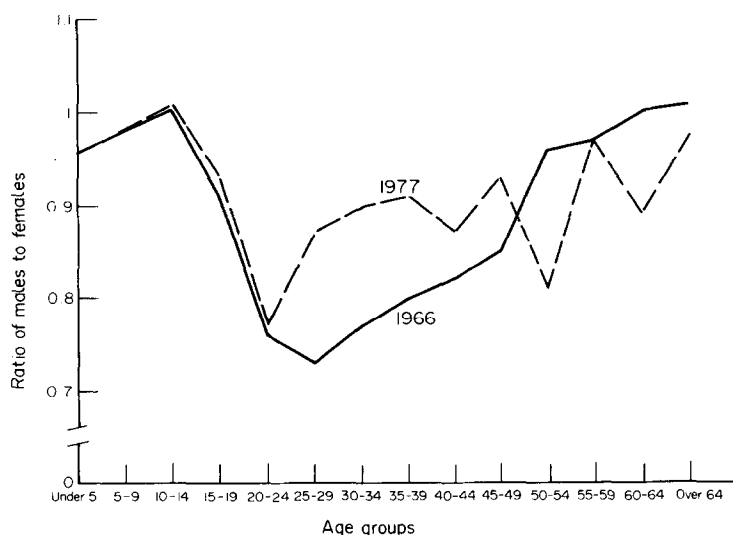


Figure 2. Sex ratios for 1966 and 1967.

Statistics required for the preparation of Figure 2 (sex ratios for 1966 and 1977)

Age group	Sex ratios	
	1966	1977
Under 5	0.96	0.96
5-9	0.98	0.98
10-14	1.03	1.06
15-19	0.91	0.93
20-24	0.76	0.77
25-29	0.73	0.87
30-34	0.77	0.90
35-39	0.80	0.91
40-44	0.82	0.87
45-49	0.85	0.93
50-54	0.96	0.81
55-59	0.97	0.97
60-64	1.00	0.89
Over 64	1.03	0.97

Source: Calculated from *Malawi Population Census, Final Reports*, 1966 and 1977.

4. CHANGES IN ECONOMIC STRUCTURE SINCE INDEPENDENCE

(a) National income accounts data

Conventionally the place to begin a discussion of development and structural change is with a consideration of the national income accounts data. For Malawi in the 1970s currently available national income accounts data is far from reliable. General criticisms of the currently available figures concern weighting

and data collection procedures.¹⁴ However, in view of our desire to identify structural change, the crucial deficiency of the national accounts is that while an estimate is made of the value of small-scale peasant production, not only of agricultural produce but also of the value of manufacturing, construction and distribution output, this estimate has become increasingly inaccurate with time. In particular, we are of the opinion that growth in the subsistence component of agricultural production has been seriously overestimated.¹⁵ Of course, the general

Table 2. *Malawi's national accounts*

(a) Composition of the monetary component of GDP 1965–1980*†				
	1965 (%)	1970 (%)	1975 (%)	1980 (%)
Agriculture, forestry and fishing	31	26	21	21
Manufacturing (including mining + quarrying)	12	17	17	15
Building and construction	4	6	6	6
Electricity, water and sanitation	1	2	2	2
Distribution	16	17	19	18
Transport, communication and storage	7	6	9	7
Banking and finance	2	2	6	7
Real estate	2	2	2	3
Community, social and personal services	25	24	19	21
Unallocable	–2	–2	–2	–5
(b) Average annual growth rates for <i>per capita</i> real GDP (subsistence and monetary), 1964–1980‡				
Year	<i>Per capita</i> real GDP (kwacha)	Average annual growth rates (%)		
1964	122.8	4.7		
1968	147.5	4.5		
1972	175.8	1.4		
1976	185.8	2.8		
1980	207.9			

* Columns may not add to 100% due to rounding. GDP estimates from 1973 onwards are provisional and are currently under review. The tentative nature of the GDP figures likely accounts for the size of 'unallocable' (see note 14).

† *Malawi Statistical Yearbook*, 1979, and *Economic Report* 1981.

‡ Calculated from *International Financial Statistics Yearbook*, 1979, and *Economic Report* 1981.

problem of inaccurate national accounts data is found in many poor countries and is a commonplace of development economics. What is perhaps surprising is that this problem is so often ignored in practice – e.g. the impressive growth rates of real GDP *per capita* shown in Table 2 are widely quoted as an index of Malawi's success in development.¹⁶ Even allowing for overestimation of growth rates, Malawi's GDP growth rate has been respectable in the context of other African countries. However, as we demonstrate later, the distribution of the benefits of this growth have been highly skewed.

The estimates for the composition of monetary GDP reported in Table 2 do not convey a very clear picture of structural change. Although it is clear that within the 'modern sector' agriculture along with community services have declined in relative importance and manufacturing and finance have increased, the size of the 'unallocable' component indicates the tentative nature of the GDP figures.

(b) Trends in output

In an account of growth of output in Malawian agriculture merits most of our attention. However, it is necessary to note the significant growth in the manufacturing sector summarized in Table 3. While this sector has exhibited strong growth, this growth is from a very small base. The strongest growth in manufacturing has been shown by import-substituting industries, but export industries (which are largely agricultural processing) has also displayed strong growth.

Trends in the volume and value of agricultural output are shown in Tables 4 and 5 which distinguish between the estate and peasant subsectors and their major crops. In the case of the peasant subsector it should be noted that these figures refer only to officially marketed peasant production (i.e. produce handled by the state marketing board) and thus subsistence and privately marketed production is excluded. This means that growth rates reported for staple

Table 3. *Annual growth rates for manufacturing and construction volume and employment and for electricity production*

Item	Period	Annual growth rate (%)
Total manufacturing volume*	1970-1980	9.4
of which:		
Food, beverage and tobacco		11.3
Footwear, clothing and textiles		4.1
Other goods		10.0
Intermediate goods for construction		4.2
Export industries		7.2
Manufacturing employment	1968-1979	7.6
Construction real value	1973-1980	6.9
Construction employment	1968-1979	6.8
Electricity production	1969-1980	11.7

Source: Calculated from *Monthly Statistical Bulletin* (February 1981), Tables 7-9.

* This growth rate is based on data calculated by the authors using current output weights. The base-weighted estimates reported in the *Monthly Statistical Bulletin* give an annual growth rate for total manufacturing volume of 8.7%.

Table 4. *Annual growth rates for volume of agricultural output, 1960-1969, 1970-1980 (calculated as an average annual growth rate from a least-squares trend line)*

Period	Estate production			Marketing Board purchases of peasant production				
	Tobacco	Tea	Sugar	Tobacco	Groundnuts	Cotton	Rice*	Maize*
1960-1969	11.2%	3.7%	—	-3.3%	6.5%	3.6%	-4.0%	21.1%
r^2	0.86	0.51	—	0.10	0.27	0.15	0.07	0.67
1970-1980	20.0%	6.6%	22.2%	3.4%	-4.8%	0.9%	5.3%	14.5%
r^2	0.94	0.90	0.92	0.13	0.25	0.04	0.29	0.62

Source: Calculated from the authors' estimates (Kydd and Christiansen, 1981b, Table 2.15 and Appendix 9).

* As noted in the text the growth rates for rice and maize do not represent trends in total marketed production of peasant origin.

Table 5. *Annual growth rates for value of agricultural output, 1960-1969, 1970-1980 (calculated as an average annual growth rate from a least-squares trend line)*

Period	Estate production			Marketing Board purchases of peasant production				
	Tobacco	Tea	Sugar	Tobacco	Groundnuts	Cotton	Rice*	Maize*
1960-1969	20.5%	2.1%	—	-0.50%	10.2%	3.1%	7.2%	—
r^2	0.69	0.24	—	0.00	0.46	0.11	0.10	—
1970-1980	31.2%	13.8%	36.8%	13.8%	12.5%	22.1%	11.3%	50.3%
r^2	0.92	0.68	0.70	0.32	0.64	0.94	0.55	0.83

Source: Calculated from the authors' estimates (Kydd and Christiansen, 1981b, Tables 2.17 and Appendix 9).

* As noted in the text the growth rates for rice and maize do not represent trends in total marketed production of peasant origin.

food crops (maize and rice) do not necessarily reflect the actual growth rates in total production or in the total marketed surplus (official and private). Since there is evidence that the share of official purchases in the total marketed surplus of staple food crops has increased in the 1970s, the actual growth rates of total mar-

keted surplus are lower than those shown in Table 4 in the cases of rice and maize.¹⁷ For other important peasant crops the growth rates are a tolerably accurate representation of actual production trends and they at best present a picture of weak growth in peasant cash crop output.

Table 6. *Number of economically active* males and females and annual growth rates, 1966–1977*

Sex	1966	1977	Average annual growth rate (%)
Males	942,092	1,266,175	2.72
Females	1,198,814	1,546,408	2.34
Total	2,140,906	2,812,583	2.51

Source: Calculated from the authors' estimates based on census data (Kydd and Christiansen, 1981b, Appendix 8).

* 'Economically active' includes all individuals over 10 years of age who are not students, full-year homeworkers or economic dependents.

In sharp contrast to the performance of peasant production, estate production has exhibited rapid growth, which has been particularly remarkable in the case of tobacco. Between the periods 1965–1969 and 1975–1979 average production levels of the major tobacco variety, flue-cured, increased by 886%. The growth rates for sugar are also very high, but this was from a small base as the Malawian sugar industry was only established in the late 1960s. Nonetheless, by 1980, a year of poor tobacco prices, sugar was Malawi's largest marketed crop by value.¹⁸ When considering the relative stability in trends in output and revenue for the two subsectors, it is worth noting that there has been much greater stability in the estate sector, as demonstrated by the r^2 -values.

(c) Trends in the structure of employment

Malawi falls into the category of countries which are experiencing rapid population growth as a result of the divergence of mortality and fertility rates. There is some evidence that Malawi's population growth rate is still increasing: the intercensal annual growth rate (1966–1977) was estimated at 2.9%, but as this contains an element of migration the underlying growth rate is thought to be about 2.6%.¹⁹

Of greater importance for the economic structure is the growth rates for economically active individuals. A summary of intercensal growth rates for economically active individuals estimated by Kydd and Christiansen (1981b) is presented in Table 6. The higher average annual growth rates for males can be attributed to the reduced opportunities for migration and the return to Malawi of many longer-term migrants (see also Figure 1). We now consider the disposition of this growing labour force. As we have shown above, in the mid-1960s the dominant sector was peasant

agriculture, which was largely based on individuals working on their own land without wage remuneration (hereafter referred to as own holding employment). The rapid growth of estate agriculture, estate associated enterprises and import substitution industries has resulted in a rapid shift out of own holding employment into wage employment. Table 7 shows the annual average growth rates for part- and full-year wage employment by sex. It is noteworthy that the proportion of economically active males engaged in either full- or part-year wage employment has doubled during the intercensal period and had reached 50% by 1977. For women, the comparable growth rates have been even higher than those for men, although it should be noted that this is from a smaller base. It is difficult to be precise about growth rates of wage employment by individual sectors. This is because comprehensive labour force data is only available for 1977, the census year, and therefore estimates of growth rates of employment by sector need to be based on quarterly labour force surveys which are far from complete in their coverage.²⁰ Table 8 presents estimates of employment levels and rates of growth by sector, based on these quarterly surveys. The nature of the sampling bias in the quarterly surveys is such that agricultural wage employment is understated to a relatively greater degree than is the case for other sectors of the economy. Despite this understatement of agricultural wage employment in the figures reported in Table 8, these nevertheless estimate that agricultural wage employment has grown faster than that for the all industries average for the period 1968–1980.²¹ The next biggest sector in terms of employment is community, social and personal services, which, however, has displayed relatively low rates of growth in employment. Manufacturing, which is the third largest employment sector, has had above average growth only in the periods 1972–1976 and 1977–1980. In 1968 the agricultural sector accounted for a

Table 7. *Wage and salaried employment by sex and duration of employment, 1966 and 1977*

Wage employment status	1966			1977			Percentage annual growth rates for wage and salaried employment		
	Percentage of total economically active *			Percentage of total economically active					
	Males	Females	Both sexes	Males	Females	Both sexes	Males	Females	Both sexes
Full-year (10-12 months)	13.0	0.9	6.2	21.1	2.5	14.5	10.5	12.0	10.7
Part-year (1-9 months)	11.0	0.8	5.3	21.0	5.5	12.5	9.0	22.2	10.8
Total full- and part-year	24.0	1.7	11.5	50.1	7.9	27.0	9.8	17.8	10.8

Source: Calculated from *Malawi Population Census 1966, Final Report* (Tables 21 and 22) and the authors' estimates (Kydd and Christiansen, 1981a, Tables 2 and 3).

* See Table 6 for definition of 'economically active'.

Table 8. *Average number of paid employees by industry for 1968, 1972, 1976 and 1980*

Industry	1968		1972		1976		1980*		Annual growth rates† (%)			
	Thousand	% of total	Thousand	% of total	Thousand	% of total	Thousand	% of total	1968-1972	1972-1976	1977-1980	
All industries	134.5	100.0	189.6	100.0	264.1	100.0	359.7	100.0	9.0	8.6	5.4	
Agriculture, forestry and fishing	44.1	32.8	63.7	33.6	103.9	39.3	182.7	50.8	9.6	13.0	5.7	
Manufacturing	17.2	12.8	23.2	12.2	36.0	13.6	39.1	10.9	7.8	11.6	6.0	
Construction	15.5	11.5	18.2	9.6	21.1	8.0	37.9	10.5	4.1	3.8	16.8	
Wholesale and retail trade, hotels and restaurants	9.4	7.0	15.9	8.4	20.7	7.8	19.5	5.4	14.0	6.8	-7.9	
Transport and storage and communication	8.1	6.0	9.8	5.2	13.0	4.9	16.5	4.6	4.9	7.3	-0.4	
Community, social and personal services	37.4	27.8	54.2	28.6	61.9	23.4	49.2	13.7	9.7	3.4	2.6	
Other‡	2.5	1.9	4.5	2.4	7.5	2.8	14.8	4.1	15.8	13.6	17.6	

Sources: *Reported Employment and Earnings Annual Report, 1972 and 1976; Economic Report 1981.*

* Data for 1980 is part of a new series which began in 1977 and was designed to increase coverage of the survey. Consequently the annual growth rates for 1977-1980 are based on the new series data, while the remaining rates are based on the old series. An overall growth rate, 1968-1980, is not presented as this would be misleading. Finally the values for 1980 are provisional.

† Annual growth rates are based on mid-year averages.

‡ Other includes mining and quarrying, electricity and water and finance and insurance.

Table 9. *Individuals working on their own holdings, * 1966-1977*

Employment group	1966			1977			Average annual growth rates (%)	
	Females (thousand)	Males (thousand)	Females as % of employment group	Females (thousand)	Males (thousand)	Females as % of employment group	Females	Males
Full-year (10-12 months)	1178.5	716.0	62.2	1423.6	631.6	69.3	2.1	-1.4
Part-year (1-9 months)	9.3	103.7	8.2	84.6	266.2	24.1	27.8	11.0
Total	1187.8	819.7	59.2	1508.2	897.8	62.7	2.7	1.0

Source: Calculated from *Malawi Population Census 1966, Final Report* (Tables 21 and 22) and authors' estimates based on *Malawi Population Census 1977, Final Report* (Kydd and Christiansen, 1981a, Tables 2 and 3).

* This refers to the estimated number of 'economically active' individuals working on their own holding. This does not include employees on peasant farms.

minimum of 44% of wage employment, by 1980 this figure was a minimum of 51%.²² There is no currently available statistical evidence which allows estimates of trends in wage employment within the peasant sector. However, given the strong evidence, which we present later, of the declining rate of return to labour engaged in peasant agriculture, it is unlikely that wage employment in this sector has increased significantly. **Thus, we believe that the increase in agricultural wage employment reported in Table 8 is almost entirely a result of the rapid expansion of the estate sector.**

The counterpart to the strong growth in wage employment has been the moderate growth of own-holding employment as demonstrated in Table 9. The most remarkable figure in this table is the annual average growth rate for male full-year employment, which shows a decline for the period. The annual growth rate for male and female full- and part-year employment on own holdings was 2.0%, which means that own-holding employment as a proportion of economically active individuals has decreased. Two features of the trends shown in Table 9 suggest that the increases in the effective application of labour to own holdings have been less than this overall 2.0% figure. Firstly, the shift to part-year employment may be interpreted as a shift to a pattern of labour time allocation whereby peasants accommodate to the labour requirements of the estates. This accommodation frequently results in an irregular pattern of wage employment of which a feature is the withdrawal of labour from own holdings at crucial periods of the peasant agricultural cycle. For this reason the contribution of part-year labour is disproportionately less than that of full-year labour. Secondly, the predominance of female labour in own-holding agriculture has been reinforced. We argue that the consequence of replacing male labour with female labour is a reduction in output per unit of labour. This is due simply to the substantial labour time females must devote to domestic non-agricultural tasks.²³

In analysing the structural changes which have occurred since Independence, we have relied largely on employment and output data rather than the national accounts. This approach makes it obvious that the key feature of structural change has been the startling growth of large-scale commercial agriculture. As in Malawi this type of agriculture is relatively labour-intensive, this trend has been associated with a transfer of labour from the peasant sector. As our historical review has shown, government intervention in the early colonial period

involved measures to curtail alternatives to estate employment for peasant labour. In Section 5 of this paper we argue that government intervention, beginning in the later 1960s, again involved measures to induce peasants to seek estate employment. In large part this was achieved by measures which eroded the viability of the substantial peasant cash cropping sector which had developed in the later part of the colonial period.

5. A MODEL OF STRUCTURAL CHANGE SINCE INDEPENDENCE

In this section we present a model of the means by which government intervention has resulted in the expansion of the large-scale commercial agriculture to a position of dominance in Malawi's agricultural sector. We argue that to acquire the financial resources needed for the development of the estate sector it was necessary to impose a heavy tax on peasants through the state crop marketing board (ADMARC). The result of this onerous tax burden was to reduce the real rate of return to peasant labour. The reduction in the real rate of return to labour in the peasant sector was sufficient to induce a transfer of labour from peasant agriculture to estates, despite a continuous decline in the real wage paid to estate labour. Finally, we argue that a consequence of the reduction in the real rates of return in peasant farming and estate employment has been to increase distributional inequity and to reduce the living standards and prospects of the majority of the population.

In order to develop the estate sector four types of resources were needed, land, labour, financial capital and technical/managerial skills. Land was obtained by the preferential allocation to estate agriculture of much of the rapidly diminishing supply of uncultivated arable land in Malawi.²⁴ The problem of acquiring technical/managerial skills was tackled by employing expatriate managers, many of whom had been employed in the Rhodesian tobacco industry. For various reasons, this solution proved less than satisfactory.²⁵ Of the remaining two factors required from the development of the estates, labour and financial resources, the most difficult to obtain in the later 1960s was financial resources. At that time there were few domestic sources of financial surplus. The commercial banking sector was unable to provide sufficient funds due to a low saving rate and a significant government subsidy was not possible because the government's immediate priority

Table 10. ADMARC net profits (loss) on crop trading at 1980 prices* (1971/1972-1978/1980) (thousand kwacha)†

Crop	1971/1972	1972/1973	1973/1974	1974/1975	1975/1976	1976/1977	1977/1978	1978/1979	1979/1980	Total profit (loss) per crop (9 years)	Total average annual profit (loss) per crop (1971/1972-1979/1980)
Tobacco	12,795 (426)	8931 (344)	5351 (313)	9061 (1054)	17,623 (45)	24,135 (1657)	36,996 (1143)	5656 (916)	3191 (1837)	123,199 (5001)	13,689 (556)
Rice											
Maize	4864 (97)	5066 (154)	3697 (3294)	1991 (410)	2103 (4758)	9181 (2470)	3638 (3336)	2969 (4463)	4412 (4929)	37,921 (16,503)	4213 (1834)
Groundnuts	1852 (1323)	2156 (3076)	3044 (2077)	5825 (2911)	767 (889)	2933 (2620)	2019 (2006)	1620 (715)	544 (1298)	20,760 (4456)	2307 (495)
Cotton											
General produce											
Total profit (loss)	17,665	12,039	13,622	15,430	16,579	34,742	40,180	5581	83	155,920	17,324
at 1980 prices											

Source: Calculated from ADMARC Crop Accounts 1972-1980.

* Prices are calculated using the GDP deflator.

† An approximate exchange value of the Malawian kwacha for this period is US\$1.2.

Table 11. *Summary of ADMARC's net assets and portfolio composition in 1978*

Composition of net assets, 1978	
Asset type	Value (K)
Fixed assets*	19,608,499
Investments and loans†	54,104,367
Designated investments‡	9,000,000
Net current assets§	38,338,949
	121,051,815
Composition of investments and loans, 1978	
Item	%
Investment in industries primarily related to smallholder agriculture	4.3
Investment in industries primarily related to estate agriculture	1.7
Investment in estate tobacco production	50.9
Investment in estate sugar production	15.1
Investment in commercial banks, investment banks and finance companies	5.4
Other industrial and commercial investment	22.6

Source: Calculated from ADMARC Accounts 1978.

* 'Fixed assets' represents the book value of plant, equipment, vehicles, land, buildings etc., less depreciation.

† 'Investments and loans' are long-term investments.

‡ 'Designated investments' relate to the crop reserve fund, which is invested entirely in Malawian government securities.

§ 'Net current assets' represents the value of crops, stores, short-term investment, cash etc., less current liabilities.

was to phase out the British budgetary grant-in-aid.²⁶ The one source of financial resources which was available was the not uncommon solution of extracting a surplus from the peasant sector via government marketing boards. This device was used by many colonial administrations in Africa and has continued to be used in varying degrees up to the present. However, as we explain later, in Malawi the policy adopted in the later 1960s of using ADMARC surpluses to finance estate development resulted in the imposition of a much heavier tax burden on the peasant sector than had been experienced in the previous decades. Table 10 summarizes ADMARC's net profits on crop trading in constant 1980 prices for the financial years 1971/1972–1979/1980. It should be noted that ADMARC purchases crops exclusively from the peasant sector, the estate sector being permit-

ted to deal directly with international buyers. It is clear from Table 10 that the most profitable crop for ADMARC, in every year, has been tobacco, with the balance of the profits being contributed by trading in groundnuts and cotton. Figure 3 shows the disparity between the trend of prices paid to the grower of dark fired tobacco (the principal peasant-grown tobacco variety) and that received by ADMARC at auction. It is clear that the price differential began to widen in the later 1960s when ADMARC failed to pass on to the peasant grower any significant proportion of world price increases. The average gross profit per annum was K20.2 million, which amounts to a gross resource extraction from the peasant sector of K181.9 million over the 9-year period.²⁷ A portion of these resources, K26.0 million, have been used to subsidize the consumption of staple foods (maize, rice and general produce). These subsidies are represented in Table 10 as the losses earned in trading in these food crops. The chief beneficiaries of these food subsidies have been estate and urban employers. However, the bulk of ADMARC's surplus has been devoted to investments in the economy. Table 11 presents an analysis of ADMARC's investment portfolio as of 1978. In large part, the categories 'designated investments' and 'investments and loans' represent the disposition of the crop trading surplus, while the categories 'fixed assets' and 'net current assets' are principally operating costs. The bulk of the surplus is accounted for by the 'investment and loans' category. The analysis shows that 66% of this category is represented by direct investment in estate production. It is noteworthy that only 4.3% has been placed in investments related to the development of peasant agriculture. Clearly, the government has used ADMARC surpluses to achieve the development of estate agriculture. In addition, ADMARC's investment in the financial sector (5.4% of total investments and loans) gave ADMARC, along with other Malawian interests, control of the country's two commercial banks. This national control of the commercial banking sector proved a key element in the general strategy of channelling financial resources to estate agriculture. This aspect of the financing of the estates only became significant in the mid-1970s. For the period 1965–1972 the average share of advances to agriculture as a percentage of total commercial bank advances was 13%, whereas in 1973 this figure was 20%, in 1977 it was 35% and by 1980 it had risen to 53%.²⁸ It should be noted that these funds were loaned out at the direction of the government, with scant

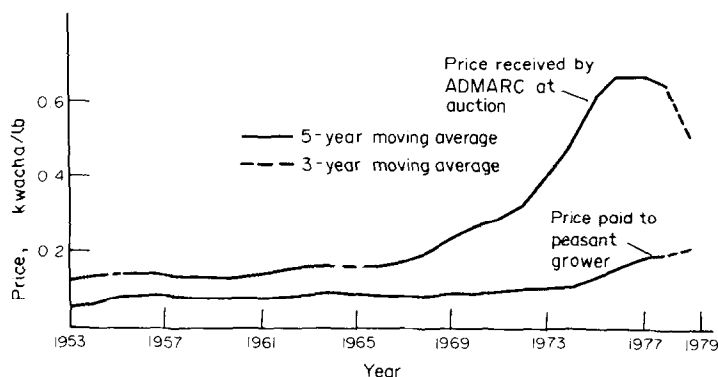


Figure 3. Price received by ADMARC at auction and price paid to peasant grower for dark fired tobacco: 5-year moving averages.

Statistics required for the preparation of Figure 3 (auction and peasant grower prices for dark fired tobacco: 5-year moving averages)

Year	5-year moving average of price received by peasant growers (kwacha/lb)	5-year moving average of price received by ADMARC at auction in Malawi (kwacha/lb)
1953	0.062	0.126
1954	0.067	0.135
1955	0.076	0.138
1956	0.081	0.141
1957	0.081	0.135
1958	0.077	0.129
1959	0.076	0.133
1960	0.078	0.135
1961	0.075	0.139
1962	0.078	0.155
1963	0.086	0.161
1964	0.090	0.161
1965	0.089	0.152
1966	0.084	0.153
1967	0.082	0.172
1968	0.081	0.192
1969	0.088	0.234
1970	0.089	0.268
1971	0.096	0.290
1972	0.101	0.325
1973	0.106	0.405
1974	0.114	0.491
1975	0.130	0.610
1976	0.156	0.653
1977	0.187	0.661
1978	0.194	0.637
1979	0.208*	0.500*

* 3-year moving average.

Source: Calculated from *Compendium of Agricultural Statistics*, 1977, updated as necessary.

regard to standard lending criteria. Most of the new estate companies formed in the mid-1970s were capitalized entirely from commercial bank loans. With poor or indifferent management, a

highly geared financial structure and interest rates in the range 13–15%, the majority of these companies have failed to establish a satisfactory financial performance in the sense that

they have been unable to meet loan interest charges and thus have increased their indebtedness to the banks. The jump in agricultural advances as a proportion of total lending from 1977 to 1980 represents, in the main, the increased indebtedness of existing operations rather than investment in new capacity. Although greatly improved tobacco prices in 1981 have eased the financial crisis which had developed in the estate tobacco industry by 1980, it seems unlikely that, in the longer run, the bank loans which have been advanced to the industry will, on aggregate, yield more than a very low rate of return.²⁹ Thus the government's determination to promote tobacco estate development at all costs has caused a misallocation of financial resources which must adversely affect the prospects for the large-scale commercial farming sector as a whole. The longer-run viability of the commercial farming sector depends on its ability to diversify and this in turn depends on the availability of capital (principally to finance irrigation development). However, the commercial banks have lost money in supporting estate development and have been forced to retrench to a position whereby they are unable to support significant diversification.

The consequences of the government's estate development policy for labour allocation between the peasant and estate sectors can best be thought of in the context of its impact on the rate of return earned by peasant labour. Table 12 shows figures for the real value per worker of ADMARC purchases from the peasant sector. It is clear from the calculated growth rates and r^2 -values that in real terms payments per worker have remained constant. There is, however, good reason to argue that average returns to labour in peasant agriculture have declined significantly. We advance three pieces of evidence for this proposition. Firstly, the costs of purchased inputs other than labour, e.g. fertilizer and insecticides, have increased much faster than output prices.³⁰ These cost increases, coupled to the relatively constant real value of cash crop payments per worker and the lack of evidence of significant increases in physical productivity per worker, clearly indicate a decrease in the real value of returns per worker. Secondly, a major non-purchased input, land, has become increasingly scarce for most peasant households. Although households do respond to land scarcity with changes in their farming systems, it seems unlikely that changed farming methods will have compensated fully for land pressure. Thirdly, as demonstrated earlier, there has been a transfer of full-year male labour from own-holding agricul-

Table 12. *Value of ADMARC purchases per worker in own-holding agriculture,* 1964–1980, at 1980 prices†*

Year	All produce including maize (kwacha)	All produce excluding maize (kwacha)
1964	9.5	8.3
1965	16.1	15.2
1966	16.7	14.5
1967	20.4	15.9
1968	9.6	6.3
1969	11.9	9.8
1970	11.3	11.0
1971	15.0	13.8
1972	16.7	14.5
1973	13.0	10.9
1974	13.8	11.5
1975	11.4	10.5
1976	15.1	12.8
1977	15.2	12.3
1978	16.4	12.9
1979	13.2	11.1
1980	11.6	9.1
Annual rate‡ of growth (%)	—0.2	—0.5
r^2	0.01	0.01

Source: Calculated from *Monthly Statistical Bulletin* (February 1981), *Yearbook of Statistics*, 1979, and *Economic Report* 1968.

* Rate of change of labour force in own-holding agriculture calculated from Kydd and Christiansen (1981b), Table 2.13, and taken as representing 1964–1980.

† Figures rendered in 1980 prices using GDP deflator. Use of the low-income consumer price index for years in which it is available gives similar results.

‡ Annual rate of change of a least-squares trend line.

ture to wage labour in the estate sector sufficient to cause an absolute decrease in male full-year employment on own holdings. Most significantly, this transfer took place in the face of decreasing real wages in the estate sector. The evidence for this decrease is the fact that for the period 1970–1980 nominal earnings per worker in estate agriculture increased at an average rate of 3.9% per annum, while the low-income retail price index increased at an average rate of 9.8% per annum: thus real earnings decreased at an annual rate of about 6%.³¹ The fact that such large numbers of workers were prepared to move into wage employment when such employment offered falling real wages suggests a marked decline in the returns to their own farming activities. Finally, we would emphasize that we are attempting to generalize about the peasant sector as a whole and that we are aware that there are limited areas which have prospered.³² However, the

evidence we have discussed suggests that falling returns have been experienced by both richer and poorer peasant groups. The observation that material input costs have risen faster than output prices is primarily of relevance to the more market-oriented peasant households. On the other hand, the observation concerning the conditions of labour transfer is revealing of the circumstances of poorer peasant farmers.

The decline in the real value of returns to peasant labour, which we have documented, is clearly an issue of humanitarian concern, but it is also crucial to an understanding of the nature of the structural changes which have taken place in the Malawian economy. We argue that the structural change which has occurred in agriculture is a result of an explicit government policy implemented primarily through control of the prices received by the peasants for their output. The government, operating through ADMARC, has kept the price paid for smallholder output artificially low and as a result has greatly reduced the net rate of return to smallholder labour. Output prices are not, of course, the only parameter influencing peasant resource allocation. In Malawi, however, they acquire particular significance because they have been most amenable to government control.

6. PROSPECTS FOR THE FUTURE AND IMPLICATIONS FOR PUBLIC POLICY

In this paper we have attempted to document and model the structural changes which have occurred in the Malawian economy since Independence. We argue that there has been a strong bias in government policy towards the development of large-scale agriculture. In analysing this large-farm bias in policy we have concentrated on what we have identified as the most important interventions by the government to channel resources to the estates. While we believe our analysis to be moving in the right direction, we recognize that it is far from comprehensive in either its scope or depth of treatment of the evidence and we hope that further research will develop points left untouched here. Despite the brevity of our analysis we believe we are in a position to comment on prospects and policies for the future.

Our analysis has shown that a large-farm-biased development strategy has led to rapid growth in estate agriculture, and, indirectly, in manufacturing. However, the majority of Malawi's people have not benefited from this development strategy and we find it difficult

to be optimistic about their prospects under this strategy. In assessing prospects it is necessary to note that since Independence wage employment and output has shown strong growth. This growth can, in large part, be attributed to four factors. Firstly, the fact that the government was able to extract substantial financial resources from the peasant sector and that these resources were used to develop estates and acquire control of the commercial banks. A second factor was the easy acquisition by estates of much of the remaining uncultivated arable land. Thirdly, there existed adequate supplies of inexpensive labour, a condition which was promoted by government policy. Finally, in the 1970s, Malawi's two most important estate crops, tobacco and sugar, experienced relatively favourable market conditions.

Looking to the 1980s, it seems improbable that a sufficient number of the aforementioned conditions can be perpetuated in order to bring about the rate of economic growth necessary to provide employment for a rapidly growing labour force, let alone achieve a general improvement in living standards. We argued above that the development strategy pursued since Independence has been damaging to the peasant sector and as a result we believe that, given land scarcity and slow technical progress, a crisis of viability in the peasant sector may not be far away. Thus the potential for further resource extraction from the peasant sector is likely to decline rapidly if present policies are continued. Further, much of the newly created estate sector has exhibited a weak financial performance and thus is unable to support additional expansion through internally generated funds. The second condition of estate expansion, available land, has changed as a result of the land frontier being reached in most parts of the country. In this regard, future output growth will need to be achieved through more intensive land use, which is likely to involve higher levels of capitalization, e.g. irrigation. However, given our conclusion concerning future sources of financial capital, capitalization will only proceed slowly. Finally, the continuation of relatively favourable market conditions for tobacco and sugar are uncertain. Profitable expansion of the sugar industry depends on the negotiation of enlarged US and EEC import quotas for sugar. At the same time, the longer term prospects for the tobacco industry are clouded. Given this prognosis, it seems unlikely that in the 1980s wage employment opportunities will be able to absorb many of the additional workers joining the labour force.

Finally, the failure of the current development strategy to achieve a broad based improvement in living standards and the unlikelihood of it achieving future success in this respect necessitates a reconsideration of the policy bias towards large-scale agriculture. We believe that a broad based and effective development strategy for Malawi must be based on prosperous and productive peasant agriculture. In Malawi, the key element in a strategy to revitalize peasant agriculture must be a price policy which significantly increases the rate of return to labour employed in peasant agriculture. Higher real prices for peasant output will not only have the short-term effect of leading

to higher output under existing production techniques, but, in the longer term, will create an environment conducive to increased capitalization and the acceptance of technical innovations. We believe that it is largely because of the government's peasant producer price policy in the last decade that the results of Malawi's rural development programme have been so disappointing.³³ We recognize that an emphasis on peasant development via higher real prices for output would damage the long-run interests of some politically powerful groups in Malawi. However, it is our view that genuine economic development will not take place in Malawi without such a policy.

NOTES

1. Other recent work on the Malawi economy includes Gupta (1974), Morton (1975) and Williams (1978). Morton's book focuses on the detail of the aid relationship between the British and Malawian governments in the period 1964–1973, during which Malawi was heavily dependent on Britain for budgetary assistance. Williams's book focuses on political history and the political and economic background to events at Independence. He does not deal in any depth with the structural evolution of the economy since Independence. Gupta's article analyses the rapid growth of manufacturing output in Malawi in the period 1964–1973. Gupta argues that the principal determinants of expansion in manufacturing are 'the importance attached to agricultural expansion and the related policy of wage restraint, under which the government and trade unions do not unduly push the general level of wages above their supply price, which is agricultural earnings' (Gupta, 1974, p. 1).

2. Acharya (1981a) defines 'market-oriented' development strategies, in contrast to 'étatist' strategies, as having the following characteristics: '(i) relatively greater reliance on markets, especially international market forces, for guiding investment and output allocation, with state enterprises limited *mainly* to the traditional public utilities and various types of infrastructure which support the directly productive sectors; (ii) slower Africanization of high/middle-level manpower; (iii) a greater willingness to let the pace and character of industrialization be governed by market forces; and (iv) a tendency to view external trade and investment as a source of gains' (Acharya, 1981a, p. 118).

3. On the point of the relationship between income distribution and development strategy Acharya (1981a) does not provide a conclusion. However, his contrast of studies of 'étatist' states (Tanzania and Ghana) and 'market-oriented' states (the Ivory Coast and Kenya) suggests that he tends to the view that the latter strategy leads to a more equitable distribution of income.

4. Acharya (1981b), p. 16.

5. Acharya (1981b), p. 18.

6. McCracken (1980), p. 7.

7. McCracken (1981), p. 3, quotes a 1906 despatch to the Colonial Office from S. Simpson, an expert on cotton growing with a year's experience in Nyasaland. 'Coffee prices fell, crops failed, when suddenly tempted by the high price of cotton everyone rushed into big acreages under this product. It was a general belief that no experience was essential to its cultivation and as to the most suitable varieties for the different situations, that point does not appear to have been considered at all. . . . All and sundry received large grants for putting in cotton and no discrimination as to the capabilities of the applicants appears to have been practiced. In a country like this at least 75% of the planters have no experience of agricultural work whatever, and would absolutely starve in most countries.' The parallels with the 1970s' estate tobacco boom are striking.

8. The estate sector grows food crops for consumption by its workers but overall is always an important food purchaser. Reasons for the food deficit of the estate sector are as follows: (i) it is a rational management strategy to concentrate supervision and labour on the higher value crop; (ii) theft by workers is difficult to control in the case of food, but is unlikely to occur in the case of tea and tobacco; (iii) a lack of machinery for drying 'wet' maize and thus the maize has to be dried in the field (the hybrid maize grown by estates is particularly vulnerable to pest attack at this stage); (iv) even with the rising value of maize relative to other crops in the 1970s, factors (i) and (ii) have meant that commercial banks have been unwilling to advance funds for commercial production of maize. A substantial project is currently being implemented to increase capacity for automated processing and storage of maize with the aim of making maize an attractive crop for the large-farm sector.

9. McCracken (1980), pp. 11–12.
10. Boeder (1974), Chap. 1.
11. The often quoted figure for 1966 of 6.3% of the *de jure* population being abroad, which was estimated from a question in the 1966 Census, is not reported in Table 1. Boeder (1974), Chap. 6, has shown convincingly that this figure is a gross underestimate. Boeder's own figures for 1972 are derived from sources in recipient countries.
12. This very approximate estimate is based on the assumption that 170,000 of 239,000 migrants returning to Malawi over the period 1972–1976 (1977 Census, Table 4.12) had left Malawi prior to 1972 and thus were included in Boeder's 1972 estimate of Malawians abroad.
13. This attempt at curtailment of international labour migration, unlike earlier attempts, was relatively successful. Prohibitions on recruitment for the South African mines were in effect from the early 1920s to 1936 and during the Second World War. The Malawian government's 1974 ban on labour recruitment was officially stated to be in response to the Francistown air disaster in which 74 Malawian miners returning from South Africa died [Mine Labour Organisation (WENELA) Ltd, 1975]. This official concern over the loss of 74 lives is somewhat hypocritical in light of the death-rate for migrant mine workers in South Africa. Official figures [Mine Labour Organisation (WENELA) Ltd, 1976] for deaths among migrant mine workers in South Africa for the years 1974 and 1975 are:

	Death-rate per 1000 per annum	
	1974	1975
From disease	1.76	1.62
From accident (includes assault and accidents not due to mine employment)	2.53	2.38

Applying an average of these death-rates to the 123,250 Malawians employed in the South African mines at the end of 1973 we infer that 528 died during the year. It should be noted that mine recruits are subject to rigorous physical examination before employment.

14. The government agency responsible for the national income accounts, the National Statistical Office, recognizes the shortcomings of the data and is presently revising the accounts. Preliminary indications are that these revisions will result in significant reduction in estimated growth rates over the period 1973–1981.

15. The National Statistical Office (NSO) estimates of subsistence production show an annual average in-

crease of over 2% per annum. These estimates are for a period during which male full-year participation in own holding agriculture actually decreased [see Section 4(c)]. At the same time, in most areas of the country there was a significant decrease in the amount of land available per peasant worker. Consequently, the implication of NSO's estimates is that there have been substantial increases in the labour productivity of peasant production of subsistence crops. We believe that there is little evidence to support such a hypothesis. A final point to indicate the importance of subsistence production estimates for overall GDP growth estimates can be made by noting that in 1972 37% of GDP was estimated to be subsistence production.

16. A case in point being Acharya's (1981a), pp. 118–119, highly favourable comments on Malawi's economic performance on the basis of reported GDP growth rates. At no point is the reader cautioned concerning inaccuracies in Malawi's GDP estimates, although this qualification is included in Malawian government documents.

17. We believe that purchases of staple food by ADMARC have been growing at a significantly faster rate than the total marketed surplus. Hence ADMARC has been taking an increasing share of the total marketed surplus. The evidence for this proposition is, firstly, the fact that ADMARC purchases have grown at a rate faster than any plausible rate of growth of demand for staple food. If it is accepted that, since independence, real *per capita* incomes for the majority of the population have at best remained broadly constant, then it seems unlikely that the rate of growth of market demand has been significantly greater than that of wage employment, which was 10.7% per annum for the period 1966–1977. Secondly, the trend for the quantity of net maize exports has been negative since the late 1960s. Beginning in 1976, Malawi became a net maize importer (Kydd and Christiansen, 1981b, p. 94). This trend in net maize exports suggests that growth in total marketed output (ADMARC plus private dealers) has been below that of market demand for staple food.

18. Kydd and Christiansen (1981b), p. 41. It should be noted that while sugar was the largest crop by value, a significant quantity of sugar is consumed domestically and thus the value of direct foreign exchange receipts for tobacco exceeded that for sugar in 1980.

19. *Malawi Population Census 1977, Final Report*, p. xvii.

20. For the years 1968–1976 this data was based on a survey of firms employing in excess of 20 employees and reported in the *Reported Employment and Earnings Annual Report*, 1970–1976. From 1977 employment data has been reported in a new series based on a wider coverage which included a sample survey of establishments employing less than 20 employees. The new series estimates, however, remain considerably lower than those estimated by the 1977 Census.

21. These growth rates have been calculated from the *Reported Employment and Earnings Annual Report* series and are summarized in Table 8.

22. We describe these figures as minima because of the relatively greater underestimation of agricultural wage employment.

23. Kydd (1978) reports the results of a labour activity survey carried out in Lilongwe district, an area of relatively developed cash cropping. It was found that female household heads and wives spent about two-thirds as much time as male household heads in field operations over the growing season. (The data were obtained by daily visits to a random sample of 120 households.) It is likely that in areas where cash cropping is less developed, female input to field operations as a proportion of male input would be higher than is the case for Lilongwe and in some areas the female input could well exceed the male input in absolute terms. Thus our argument that the increased feminization of the peasant labour force will lead to reduced output per worker depends on an 'educated guess' about the role of male labour prior to their withdrawal from work on their own holdings, and thus requires empirical validation.

24. The Malawian Ministry of Agriculture has estimated that in 1977 88% of potentially arable land was under cultivation and air photo interpretation for 1971 and 1975 indicated that the cultivated area was increasing at a rate of 3.5% per annum. A regional breakdown shows that two-thirds of the remaining land is located in the Northern Region, where only 12% of the population lives. Thus in the populous Central and Southern Regions the land frontier has effectively been reached. In 1979, 14% of the arable land under cultivation was in the estate sector. Although figures for earlier years are not available, it is obvious on the basis of output growth that the area under estate cultivation has been expanding at a much faster rate than that under peasant cultivation. Most of the land for new estates has been acquired by senior politicians, public servants and military and police officers using their influence to secure the remaining uncultivated arable land in their home areas. Generally, this has not involved eviction of peasants, but has further constrained land availability for peasants.

25. The source of supply of managers was largely Rhodesia. Until the final years of the war for independence in Zimbabwe, employment in Malawi was not a very attractive option for these managers and thus Malawi experienced difficulty in recruiting competent managers. Further, management problems were exacerbated by a bonus system which imposed a short-term planning horizon on managers to the detriment of the interests of the estates. An additional factor has been the government's failure to initiate a programme to provide Malawians with the necessary skills to assume managerial positions in the tobacco industry.

26. During the federal period Malawi received a substantial subsidy from the federal government. Thus,

at Independence, the Malawian government had obligations far exceeding its ability to meet them. This deficit was financed by a British budgetary grant-in-aid over the period 1964–1972. The priority accorded to the elimination of this assistance required a policy of budgetary stringency allied to an expansion of the tax base. For further details of British assistance in this period see Morton (1975).

27. An approximate distribution of this profit per worker (full- and part-year) in the peasant sector is as follows. In 1977 there were 1.945 million workers in the peasant sector. ADMARC's gross extraction in 1977/1978 was 44.7 million kwacha. This results in a figure for gross resource extraction of 23 kwacha per worker per annum. As a rough guide to the significance of this 23-kwacha figure, it represents at least a third of the average *per capita* incomes of the poorest 90% of the population.

28. Calculated from data in *Reserve Bank of Malawi Financial and Economic Review*, Vol. 12, Table 1.7.

29. Commercial bank borrowing has been much more expensive than loans from ADMARC and, as we have noted, high loan charges have been an important factor in the failure of these companies to establish a satisfactory financial performance. Even so, commercial bank lending to the estates contains a significant, and increasing, element of subsidy. Firstly, at the direction of the government, agriculture was exempted from a 2% increase in commercial bank lending rates in 1979. Secondly, in the face of a deteriorating balance of payments the government has resorted to rationing credit by sector to control money supply growth. In the implementation of this policy agriculture has received very favourable treatment and the commercial banks have been encouraged to continue to support clearly non-viable tobacco companies.

30. By the later 1970s peasant farmers were purchasing significant quantities of material inputs. For example, fertilizer use in the peasant sector expanded at an annual rate of 13.5% over the period 1967–1976 and in 1976 (the latest year for which figures are available) we calculate that average fertilizer use per household practising own-holding agriculture (including those with resident part-year wage earners) was about 50 lb.

31. Calculated from *Malawi Statistical Yearbook*, 1979 and *Malawi Statistical Bulletin*, December 1980. These figures are inclusive of estimated benefits in kind.

32. For example, participants in irrigated rice schemes.

33. An analysis of the experience of Malawi's small-farmer development programmes up to the later 1970s can be found in Kydd and Christiansen (1981b), pp. 92–112.

REFERENCES

- Acharya, S. N., 'Perspectives and problems of development in low income sub-Saharan Africa', World Bank Staff Working Paper No. 300 (Washington: World Bank, 1978), pp. 1-64.
- Acharya, S. N., 'Perspectives and problems of development in sub-Saharan Africa', *World Development*, Vol. 9, No. 2 (1981a), pp. 109-147.
- Acharya, S. N., 'Development perspectives and priorities in sub-Saharan Africa', *Finance and Development*, Vol. 18, No. 1 (1981b), pp. 16-24.
- Boeder, R. B., 'Malawians abroad: the history of labour emigration from Malawi to its neighbours, 1899 to the present day', Ph.D. (Michigan State University, 1974).
- Gupta, D., 'Industrial development in Malawi', *Journal of Social Science*, University of Malawi, Vol. 3 (1974), pp. 35-47.
- Humphrey, D. H., 'Malawi's economic progress and prospects', *Eastern Africa Economic Review*, Vol. 5, No. 2 (1973), pp. 71-104.
- Kydd, J. G., 'Labour allocation and crop labour requirements, 1969/70', Lilongwe Land Development Programme, Evaluation Unit, Farm Management Survey 1969/1970-1971/1972 (Lilongwe: 1978).
- Kydd, J. G. and R. E. Christiansen, 'The distribution of income in Malawi in 1977', University of Malawi, Centre for Social Research, Income Distribution Project, Working Paper No. 1 (Zomba: 1981a).
- Kydd, J. G. and R. E. Christiansen, 'Structural change and trends in equity in the Malawian economy, 1964-1980', University of Malawi, Centre for Social Research, Income Distribution Project, Working Paper No. 2 (Zomba: 1981b).
- Malawi Government, *Economic Report 1981* (Zomba: Government Printer, 1981).
- Malawi Government, National Statistical Office, *Monthly Statistical Bulletin*, monthly series, 1968-1981 (Zomba: National Statistical Office).
- Malawi Government, National Statistical Office, *Reported Employment and Earnings Annual Report, 1970-1976* (Zomba: Government Printer, 1970-1976).
- Malawi Government, National Statistical Office, *Malawi Population Census 1966, Final Report* (Zomba: Government Printer, 1971).
- Malawi Government, National Statistical Office, *Malawi Population Census 1977, Final Report*, Vols 1 and 2 (Zomba: Government Printer, 1980a).
- Malawi Government, National Statistical Office, *Malawi Statistical Yearbook, 1979* (Zomba: Government Printer, 1980b).
- McCracken, J., 'Production and control in British Central Africa, 1905-1940', Chancellor College History Seminar 1980/81, Paper No. 1 (Zomba: 1980).
- McCracken, J., 'Experts and expertise in colonial Malawi', Chancellor College Staff Seminar Paper No. 13 (Zomba: 1981).
- Mine Labour Organisation (WENELA) Ltd, 'Reports and financial statements for the year ended 31st December 1974' (Johannesburg: 1975).
- Mine Labour Organisation (WENELA) Ltd, 'Reports and financial statements for the year ended 31st December 1975' (Johannesburg: 1976).
- Morton, K., *Aid and Dependence: British Aid to Malawi* (London: Croom Helm, in association with Overseas Development Institute, 1975).
- Reserve Bank of Malawi, *Financial and Economic Review*, Vols 1-12 (Blantyre: 1968-1980).
- Thomas, S., 'Economic developments in Malawi since Independence', *Journal of Southern African Studies*, Vol. 2, No. 1 (1975), pp. 30-51.
- Williams, T. D., *Malawi: The Politics of Despair* (Ithaca: Cornell University Press, 1978).